

2026山东大学图论组合青年论坛

会议手册

主办单位

山东大学数学学院

山东大学数据科学研究院

中国运筹学会图论组合分会

山东 济南

2026年6月5日-7日

山东大学图论组合青年论坛日程安排

6月6日上午

(明德楼 C704, 腾讯会议: 661-4613-8067 密码: 369835)

时间	报告人	题目	主持人
08:30-09:00	曹梦雨	On degree bounds of k -uniform hypergraphs with bounded matching number	王 赞
09:00-09:30	池 骋	A solution to the Frankl-Kupavskii conjecture on the Erdős-Kleitman matching problem	
09:30-10:00	赵婧雯	Perfect Matchings in Random Sparsifications of Dense Hypergraphs	
10:00-10:20	茶歇		
10:20-10:50	陈耀斌	Hitting time for Hamilton cycles in pseudorandom graphs	周文玲
10:50-11:20	李路易	Rainbow pancyclicity in graph system with the Ore-type condition	
11:20-11:50	张林朋	An Ore-type condition for H -tilings in graphs	
12:00-13:00	午餐（齐园三楼）		

6月6日下午

(明德楼 C704, 腾讯会议: 661-4613-8067 密码: 369835)

时间	报告人	题目	主持人
14:30-15:00	钟嘉旭	Sufficient conditions for fractional k -factor-critical graphs with minimum degree to be k -factor-critical	孙婉婷
15:00-15:30	宋得萍	Bound on shortest cycle covers	
15:30-16:00	袁小利	Degree stability of graphs forbidding a family of odd cycles	
16:00-16:20	茶歇		
16:20-16:50	胡启明	Lower Bounds for Maximum Weight Bisections of Triangle-free Subcubic Graphs	张林朋
16:50-17:20	谭 澳	Multicolor Boolean Ramsey numbers	
17:20-17:50	刘睿龙	A double-exponential lower bound for $r_4(5, n)$	
18:00-20:00	晚餐（齐园三楼 319）		

6 月 6 日上午

(明德楼 C717 腾讯会议: 676-2587-5357 密码: 369835)

时间	报告人	题目	主持人
08:30-09:00	葛川源	Equiangular lines and eigenvalue multiplicities of graphs	江素云
09:00-09:30	张俊雪	Minimum spectral radii of saturated graphs for linear forests	
09:30-10:00	吉一喆	New arithmetic invariants for cospectral graphs	
10:00-10:20	茶歇		
10:20-10:50	连晓盼	Tight upper bound on the clique size in the square of 2-degenerate graphs	王志兰
10:50-11:20	孙 爽	On the structures of {diamond, bowtie}-free graphs that do not contain an induced subdivision of K_4	
11:20-11:50	曾嘉晟	Polylogarithmic Bounds for Nested Cycles without Geometric Crossings	
12:00-13:00	午餐（齐园三楼）		

6 月 6 日下午

(明德楼 C717 腾讯会议: 676-2587-5357 密码: 369835)

时间	报告人	题目	主持人
14:30-15:00	郝常清	Color domination versus packing in graphs	朱佳露
15:00-15:30	匡珮汝	Nearly tight bound for rainbow clique subdivisions in properly edge-colored graphs and applications	
15:30-16:00	邓成龙	List colouring of toroidal grids	
16:00-16:20	茶歇		
16:20-16:50	杨宁远	Cayley Graph Chromatic Numbers and Applications	聂向向
16:50-17:20	周 欢	Fractional balanced chromatic number and arboricity of planar graphs	
17:20-17:50	李 吉	Sharp bounds and characterizations for cut-component numbers of graphs	
18:00-20:00	晚餐（齐园三楼 319）		

6 月 7 日上午
(明德楼 C704, 腾讯会议: 661-4613-8067 密码: 369835)

时间	报告人	题目	主持人
08:30-09:00	郭姗姗	3-Decomposition Conjecture of cubic graph	欧阳前程
09:00-09:30	陶裘祯宇	A structure theorem for rooted connectivity in bidirected graphs	
09:30-10:00	喻俊燃	Maximum size of digraphs with prescribed k -arc-strong connectivity	
10:00-10:10	茶歇		
10:10-10:40	张灵浩	Hamilton cycles in oriented graphs with mild connectivity constraints	蔡俊青
10:40-11:10	周敬荣	Ore-type degree condition for Hamiltonicity and pancyclicity of hypergraphs and bipartite graphs	
11:10-12:00	问题分享、自由讨论		
12:00-13:00	午餐（齐园三楼）		

6 月 7 日下午
(腾讯会议: 661-4613-8067 密码: 369835)

时间	报告人	题目	主持人
14:00-14:30	黄欣祺	On the blowup threshold of graphs	王光辉
14:30-15:00	舒希超	Fundamental cycles in grid graphs	
15:00-15:30	高 峻	On cliques in hypergraphs	
15:30-16:00	程杨杨	1-factorization of k -uniform hypergraphs under codegree condition	
16:00-16:30	孙淑敏	Rational codegree Turán density of hypergraphs	颜 谨
16:30-17:00	王岚超	Monochromatic Matchings and Tilings	
17:00-17:30	阎知非	Triangle packings in randomly perturbed graphs	
17:30-18:00	刘 苗	A Sharp Product Bound for r -Cross-Intersecting Families	

报告摘要

On degree bounds of k -uniform hypergraphs with bounded matching number

曹梦雨

中国人民大学

We study the connection between the degree sequence of a k -uniform hypergraph and the size of its largest matching. Let $\mathcal{F}$ be a k -uniform hypergraph on n vertices and let $d_1 \geq d_2 \geq \cdots \geq d_n$ be the vertex degrees arranged in non-increasing order. For integers $k \geq 2$, $s \geq 2$ and $n > 2sk$, we prove that if the $(2sk+1)$ -th largest degree satisfies $d_{2sk+1} > \binom{n-1}{k-1} - \binom{n-s}{k-1}$, then $\mathcal{F}$ contains a matching of size at least s . This can be viewed as a generalization of theorems. Moreover, by relaxing the range of n , we obtain the same bound for the $(k+2s-2)$ -th largest degree vertex. Note that the number $k+2s-2$ is optimal.

For a k -set of vertices $S \subseteq [n]$, the degree of S is defined as $\deg(S) = \sum_{v \in S} \deg(v)$, and the minimum of $\deg(S)$ over all non-edge k -subsets $S \notin E(\mathcal{F})$ of $V(\mathcal{F})$ is the *Ore-degree* of $\mathcal{F}$, denoted by $\sigma_k(\mathcal{F})$. Balogh, Palmer and Raeisi proved: for $s \geq 2$ and $n \geq 3k^2(s-1)$, if $\sigma_k(\mathcal{F}) > k \left(\binom{n-1}{k-1} - \binom{n-s}{k-1} \right)$, then $\mathcal{F}$ contains a matching of size s . They also conjectured that the result holds when $n > sk$. As a corollary, we prove that the bound on n can be taken to be linear in sk ($n \geq 3sk$).

个人简介：曹梦雨, 中国人民大学副教授, 2020 年 6 月在北京师范大学取得理学博士学位。2020 年 9 月至 2022 年 9 月在清华大学进行博士后研究工作, 2022 年 9 月入职中国人民大学数学学院。研究兴趣为极值图论和代数组合学, 在 J. Combin. Theory Ser A, SIAM J. Discrete Math., European J. Combin. 等期刊发表多篇 SCI 学术论文, 主持国家自然科学基金一项和北京市自然科学基金面上项目一项。

Polylogarithmic Bounds for Nested Cycles without Geometric Crossings

曾嘉晟

上海交通大学

Cycles $C_1, \dots, C_k$ in a graph are called nested if $V(C_k) \subseteq V(C_{k-1}) \subseteq \dots \subseteq V(C_1)$. If, in addition, their edge sets are pairwise disjoint, then they are edge-disjoint nested cycles. Let C be a cycle with vertices $V(C) = x_1, x_2, \dots, x_m$ and edges $E(C) = \{x_i x_{i+1} \mid i \in [m-1]\} \cup \{x_m x_1\}$. A cycle C' of length ℓ on vertices $x_{i_1}, x_{i_2}, \dots, x_{i_\ell}$ with $i_1 = 1$ and $1 < i_2 < i_3 < \dots < i_\ell \leq m$ has no crossing edges if its edges are exactly

$$(x_{i_1}, x_{i_2}), (x_{i_2}, x_{i_3}), \dots, (x_{i_{\ell-1}}, x_{i_\ell}), (x_{i_\ell}, x_{i_1}).$$

A problem of Erdős asks for the minimum number $f_k(n)$ of edges in an n -vertex graph forcing k edge-disjoint nested cycles without geometric crossings in 1976. Fernández, Kim, Kim and Liu in 2022 proved that there exists a constant C such that $f_2(n) \leq Cn$. In this paper, we prove that for $k \geq 3$, there exists a constant C_k depending only on k such that

$$f_k(n) \leq C_k n (\log n)^{k-1} (\log \log n)^{k-3}.$$

The proof combines the robust sublinear expander framework of Alon, Bucić, Sauermann, Zakharev and Zamir with a controlled wrapping lemma that permits the layers to be built successively with controlled length.

个人简介：曾嘉晟，上海交通大学硕士研究生，导师为张晓东教授。

Hitting time for Hamilton cycles in pseudorandom graphs

陈耀斌

复旦大学

Consider the random subgraph process on a base graph G with n vertices: we generate a sequence $\{\mathbf{G}_t\}_{t=0}^{|E(G)|}$ by taking a uniformly random ordering of the edges of G and then adding these edges one by one to the empty graph G_0 on the same vertex set. We prove that there is a constant $C > 0$ such that if G is an (n, d, λ) -graph with $d/\lambda \geq C$, then with high probability, the hitting time for the appearance of a Hamilton cycle coincides with the hitting time for reaching minimum degree 2. This resolves questions posed by Alon–Krivelevich in 2019 and by Frieze–Krivelevich in 2002. As a consequence, we determine the sharp threshold for Hamilton cycles in (n, d, λ) -graphs with $d/\lambda \geq C$ for all d sufficiently large. Lastly, we extend our result to the minimum degree $2k$ versus k edge-disjoint Hamilton cycles setting for $k \leq c \cdot \min\{d, \log n\}$ where c is a constant depending on C . This advances on a question asked by Frieze.

个人简介：陈耀斌，复旦大学在读博士，导师为吴河辉教授。

1-factorization of k -uniform hypergraphs under codegree condition

程杨杨

帕绍大学

Let $2 \leq k < n$ be integers such that k divides n . The classical Baranyai's theorem states that every complete k -uniform hypergraph can be decomposed into perfect matchings. We prove a codegree condition of Baranyai's theorem as follows: For fixed k and $\varepsilon > 0$, the following holds for all sufficiently large n . An n -vertex k -graph G with $\delta(G) \geq (1/2 + \varepsilon)n$ can be decomposed into perfect matchings if and only if $k \mid n$ and G is vertex-regular. This generalizes Baranyai's theorem when n is sufficiently large and solves a conjecture proposed by Glock, Kühn and Osthus. The talk is based on joint work with Stefan Glock and Andrew Lane.

个人简介：程杨杨，现为帕绍大学博士后，合作导师为 Stefan Glock。研究方向为极值组合，组合设计。

A solution to the Frankl-Kupavskii conjecture on the Erdős-Kleitman matching problem

池骋

上海交通大学

For integers $n \geq s \geq 2$, let $e(n, s)$ be the maximum size of a family $\mathcal{F} \subseteq 2^{[n]}$ with no s pairwise disjoint members. The problem of determining $e(n, s)$ is closely related to its uniform counterpart, the Erdős Matching Conjecture. Frankl and Kupavskii conjectured an exact formula for $e((m+1)s - \ell, s)$ when $1 \leq \ell \leq \lceil s/2 \rceil$. We prove that for every fixed $m \geq 3$ and sufficiently large s , the extremal families for $e((m+1)s - \ell, s)$ are

$$\mathcal{P}(m, s, \ell; L) := \{A \subseteq [n] : |A| + |A \cap L| \geq m+1\}$$

for some L with $|L| = \ell - 1$ when $1 \leq \ell \leq (\frac{m+1}{2m+1} - o(1))s$. In particular, this confirms the Frankl-Kupavskii conjecture for every fixed $m \geq 3$ and all sufficiently large s . For $m = 3$ and sufficiently large s , we determine the exact range of ℓ for which $\mathcal{P}(3, s, \ell; L)$ is extremal.

Theorem 1. Fix $m \geq 3$ and $\varepsilon > 0$. There is an integer $s_0 = s_0(m, \varepsilon)$ such that the following holds for all $s \geq s_0$. Suppose

$$1 \leq \ell \leq \left(\frac{m+1}{2m+1} - \varepsilon\right)s, \quad n = (m+1)s - \ell.$$

If $\mathcal{F} \subseteq 2^{[n]}$ satisfies $\nu(\mathcal{F}) < s$, then $|\mathcal{F}| \leq |\mathcal{P}(m, s, \ell)|$. Moreover, equality holds if and only if $\mathcal{F} = \mathcal{P}(m, s, \ell; L)$ for some $L \in \binom{[n]}{\ell-1}$. Let $n = 4s - \ell$ for some ℓ with $1 \leq \ell \leq s$ and $L' \in \binom{[n]}{3\ell-1}$.

Define

$$\mathcal{P}'(s, \ell; L') = \binom{L'}{3} \cup \binom{[n]}{\geq 4}.$$

Let $t(s)$ be the root of $|P(3, s, \ell)| = |P'(s, \ell)|$ with $t(s) > 1$. A direct calculation gives

$$t(s) := \frac{17 - 18s + \sqrt{49 - 852s + 1284s^2}}{20} = \frac{\sqrt{321} - 9}{10}s + O(1) \approx 0.8916s + O(1).$$

For $m = 3$, we prove the following sharper theorem.

Theorem 2. There exists an integer s_0 such that the following statements hold for all s, ℓ with $s \geq s_0$ and $1 \leq \ell \leq t(s)$. Write $n = 4s - \ell$. If $\mathcal{F} \subseteq 2^{[n]}$ satisfies $\nu(\mathcal{F}) < s$, then $|\mathcal{F}| \leq |\mathcal{P}(3, s, \ell)|$. Moreover, when $\ell < t(s)$, equality holds if and only if $\mathcal{F} = \mathcal{P}(3, s, \ell; L)$ for some $L \in \binom{[n]}{\ell-1}$. If $t(s)$ is an integer and $\ell = t(s)$, equality holds if and only if either $\mathcal{F} = \mathcal{P}(3, s, \ell; L)$ for some $L \in \binom{[n]}{\ell-1}$ or $\mathcal{F} = \mathcal{P}'(s, \ell; L')$ for some $L' \in \binom{[n]}{3\ell-1}$.

个人简介：池骋, 上海交通大学博士后, 合作导师为汪彦副教授, 2025年毕业于华东师范大学, 导师为袁龙图副教授. 研究方向为极值图论。

List colouring of toroidal grids

邓成龙

浙江师范大学

Combinatorial Nullstellensatz is a powerful tool in the study of list colouring of graphs. To prove that a graph G has choice number $\text{ch}(G)$ at most k , it suffices to prove that G has Alon-Tarsi number $AT(G)$ at most k . Dvořák proved a strengthening of Combinatorial Nullstellensatz, which shows that the coefficients of monomials in the expansion of the graph polynomial P_G contain more information about list colouring of G than its Alon-Tarsi number. It says that if L is a list assignment of G , and $\rho : V(G) \rightarrow \mathbb{Z}$ satisfies $\rho(v) < |L_v|$ for all v and $\sum_{v \in V(G)} \rho(v) = |E(G)| - 1$ and G is not L -colorable, then for every colour c , $\sum_{v \in V(G)} [x^{\rho+1_v}] P_G \cdot \chi_{L,c}(v) = 0$. This result has the potential to prove $\text{ch}(G) \leq k$ even if $AT(G) > k$. Dvořák gave a few small example graphs for this to be the case. However, applying Dvořák's result involves complicated calculations, and it is a challenging task to apply this result to some interesting infinite families of graphs. In this talk, we apply Dvořák's result to the study of list colouring of toroidal grids. It is known that for any odd positive integers ℓ, m , $T_{\ell,m} = C_\ell \square C_m$ has Alon-Tarsi number 4. On the other hand, it is conjectured that these graphs have choice number 3. This family of graphs seems to be a perfect candidate for the application of Dvořák's result. We formulate a conjecture on the rank of a matrix whose entries are $[x^\rho] P_G$ for various monomials x^ρ of P_G , where $G = T_{\ell,m}$. Using Dvořák's result, we prove that this conjecture implies that $T_{\ell,m}$ is 3-choosable. We verify this conjecture for the case $\ell = 3$. As a consequence, the toroidal grids $T_{3,m}$ are 3-choosable for all odd integers m .

个人简介：邓成龙，浙江师范大学在读博士，导师为朱绪鼎教授，研究方向为图的染色。

On cliques in hypergraphs

高峻

华威大学

We prove that for any $k \geq 3$, every k -uniform hypergraph on n vertices contains at most $n - \omega(1)$ different sizes of cliques (maximal complete subgraphs). In particular, the 3-uniform case answers a question of Erdős.

个人简介：高峻，华威大学博士后，合作导师 Oleg Pikhuiiko 教授，2022 年博士毕业于中国科学技术大学，导师马杰教授，研究方向为极值组合。

Equiangular lines and eigenvalue multiplicities of graphs

葛川源

福州大学

A set of lines in $\mathbb{R}^d$ passing through the origin is called equiangular if every two lines have a common angle. In this talk, we will first review the connection between the eigenvalue multiplicities of graphs and equiangular lines. We will then present our recent work on eigenvalue multiplicities of graph and its applications to equiangular lines. This is based on jointed works with Shiping Liu.

个人简介：葛川源，福州大学数学与统计学院副教授。2025 年于中国科学技术大学获得博士学位，导师为刘世平教授。主要研究领域为离散几何、图论及其应用。

3-Decomposition Conjecture of cubic graph

郭姗姗

福州大学

The 3-decomposition conjecture states that every connected cubic graph can be decomposed into a spanning tree, a union of cycles, and a matching. In this talk, we show that if every vertex of a cubic graph is contained in a 3-cycle or a 4-cycle, then the conjecture holds.

个人简介：郭姗姗，福州大学博士后，合作导师为范更华教授，研究方向为子图分解。

Lower Bounds for Maximum Weight Bisections of Triangle-free Subcubic Graphs

胡启明

南京师范大学

Let $G = (V, E)$ be a graph. For $v \in V(G)$, let $d_G(v)$ be the degree of v in G . A graph is called *cubic* if every vertex has degree 3, and *subcubic* if every vertex has degree at most 3. A graph G is called *triangle-free* if it contains no K_3 .

Let V_1 and V_2 be two disjoint subsets of $V(G)$. If $V_1 \cup V_2 = V(G)$, then (V_1, V_2) is called a *bipartition* of G , and $e(V_1, V_2)$ is the size of the bipartition. If G is equipped with a weight function w defined on $E(G)$, then G is called a *w-weighted graph*. A *bisection* of G is a bipartition (or cut for weighted graph) of G which requires that $||V_1| - |V_2|| \leq 1$.

Recently, Gerke et al. proved a bound for weighted triangle-free subcubic graphs.

Theorem 1. *Every triangle-free subcubic w-weighted graph G admits a bisection of weight at least $\theta_1 w(G)$ unless $G \cong K_{1,3}$, where $\theta_1 = \frac{613}{855} \approx 0.716959$.*

We study the maximum weight bisection of weighted triangle-free subcubic graphs. To handle the balancedness constraint, we introduce the concept of a *unit* - a small block consisting of an induced subgraph with a prescribed balanced bipartition. This framework captures the “block-decomposition” approach used in both Gerke et al. and our work: by randomly and independently flipping the two sides of each unit, the problem of finding a large bisection reduces to constructing a collection of units with large $w'(B) = w(E^+(B)) - w(E^-(B))$, where $E^+(B)$ and $E^-(B)$ respectively denote edges going across or within the prescribed bipartitions of the units.

Building on this idea, we extend a key lemma of Gerke et al. from balanced bipartite subgraphs to general balanced subgraphs. This extension allows for a more flexible construction of random balanced subgraphs and leads to improved lower bounds.

Our main results are the following two theorems.

Theorem 2. *Every triangle-free subcubic w-weighted graph G has a bisection of weight at least $\theta_2 w(G)$ unless $G \cong K_{1,3}$, where $\theta_2 = \frac{18}{25} = 0.72$.*

Theorem 3. *Every $\{C_3, C_5\}$ -free subcubic w-weighted graph G has a bisection of weight at least $\theta_3 w(G)$ unless $G \cong K_{1,3}$, where $\theta_3 = \frac{39}{53} \approx 0.735849$.*

For the first result, we additionally introduce a novel method to merge two small units into a larger balanced subgraph. As a result, we prove Theorem 1. This improves the previous bound $\frac{613}{855} \approx 0.716959$ established by Gerke et al. and represents a step toward the conjecture $\frac{11}{15} \approx 0.733333$ proposed in their paper.

The second result further restricts to $\{C_3, C_5\}$ -free graphs. Applying the same extended lemma (without the merging technique), we obtain Theorem 2. This bound exceeds the conjecture for

general triangle-free graphs, showing that 5-cycles are the primary obstacle to achieving a higher lower bound.

个人简介：胡启明，南京师范大学博士，导师为许宝刚教授，主要研究图划分和图染色问题。

On the blowup threshold of graphs

黄欣祺

中国科技大学-韩国基础科学研究院

The chromatic threshold, introduced by Erdős and Simonovits, asks when a minimum-degree condition forces every H -free graph to have bounded chromatic number. A natural strengthening is the homomorphism threshold, where one asks for a bounded H -free homomorphic image instead of only bounded colourability. These thresholds measure, in different ways, when dense H -free graphs must have bounded structural complexity.

In this talk, we discuss a further strengthening, called the blowup threshold. A graph is a blowup of a graph F if it is obtained by replacing every vertex of F by an independent set, and every edge of F by a complete bipartite graph between the corresponding parts. The blowup threshold $\delta_B(H)$ asks when every sufficiently large maximal H -free graph G with $\delta(G) \geq \alpha|V(G)|$ must be a blowup of some bounded-size graph. Thus, compared with the homomorphism threshold, the blowup threshold asks for an exact bounded-template structure rather than only a bounded quotient.

Our main results show that the blowup threshold differs from the chromatic threshold in three fundamental ways. First, although many non-bipartite graphs have chromatic threshold zero, we prove that $\delta_B(H) > 0$ for every non-bipartite graph H . Second, the set of possible blowup-threshold values is much richer: for every integer $k \geq 2$, there exists a 3-chromatic graph H_k such that $\delta_B(H_k) = 1/(2k)$. These values do not occur as chromatic thresholds of 3-chromatic graphs. Third, we prove that δ_B is not monotone even under taking induced subgraphs. This contrasts with the chromatic threshold, whose monotonicity is an important ingredient in its complete classification.

The proofs combine exact structural arguments with pseudo-blowup constructions, which produce dense maximal H -free graphs whose twin quotients are necessarily large. These results suggest that determining blowup thresholds requires new ideas beyond the existing framework for chromatic thresholds.

个人简介: 黄欣祺, 中国科技大学-韩国基础科学研究院在读博士生, 导师为刘鸿教授 (IBS)、张先得教授 (USTC)、Tuan Tran 教授 (USTC)。研究方向为极值图论、加性组合与编码理论。

New arithmetic invariants for cospectral graphs

吉一喆

南开大学

An invariant for cospectral graphs is a property shared by all cospectral graphs. In this talk, we present three new invariants for cospectral graphs, characterized by their arithmetic nature and apparent novelty. Specifically, let G and H be two graphs with adjacency matrices $A(G)$ and $A(H)$, respectively. We show, among other results, that if G and H are cospectral, then $e^T A(G)^m e \equiv e^T A(H)^m e \pmod{4}$ for any integer $m \geq 0$, where e is the all-one vector. As a simple consequence, we demonstrate that under certain conditions, every graph cospectral with G is determined by its generalized spectrum.

个人简介：吉一喆，南开大学数学科学学院博士后，合作导师：史永堂教授，博士毕业于西安交通大学，研究方向：代数图论。

Nearly tight bound for rainbow clique subdivisions in properly edge-colored graphs and applications

匡珮汝

上海交通大学

An edge-colored graph is said to be rainbow if all its edges have distinct colors. In this paper, we study the rainbow analogue of a fundamental result of Mader [*Math. Ann.* **174** (1967)] on the existence of subdivisions in graphs with large average degree. This is part of the study of rainbow analogues of classical Turán problems, a framework systematically introduced by Keevash, Mubayi, Sudakov and Verstraëte [*Combin. Probab. Comput.* **16** (2007)]. We prove that every properly edge-colored graph on n vertices with average degree at least $(\log n)^{1+o(1)}$ contains a rainbow subdivision of K_t , which is tight up to the $o(1)$ term. So it essentially resolves a question raised by Jiang, Methuku and Yepremyan [*European J. Combin.* **110** (2023)] on rainbow clique subdivisions, and also implies a result of Alon, Bucić, Sauer mann, Zakharov and Zamir [*Proc. Lond. Math. Soc.* **130** (2025)] on rainbow cycles. In addition, we present several applications of our result to problems in additive combinatorics, number theory and coding theory.

个人简介：匡珮汝，上海交通大学在读博士，导师为汪彦副教授。

Sharp bounds and characterizations for cut-component numbers of graphs

李吉

南京师范大学

System reliability is intrinsically linked to the vertex connectivity of the underlying topological structure. While the classical vertex connectivity $\kappa(G)$ measures the minimum size of vertex cut required to disconnect a graph, we investigate the cut-component number $c(G)$ in the paper, a parameter recently proposed to address the diagnostic problem under the PMC model. We first establish sharp upper and lower bounds for $c(G)$ and characterize the extremal graphs attaining these bounds. Furthermore, we introduce an efficient algorithm for computing $c(G)$, offering an improvement in time complexity over existing approaches. We also study Nordhaus-Gaddum type results for $c(G)$ over the class of graphs for which both G and its complement $\overline{G}$ are connected, deriving tight bounds for $c(G) + c(\overline{G})$ and $c(G) \cdot c(\overline{G})$. Moreover, based on these bounds, we present a refined theorem for determining the diagnosability of multiprocessor systems.

个人简介：李吉，南京师范大学在读博士生，导师为张晓岩教授，研究方向为图论和组合优化。

Rainbow pancyclicity in graph system with the Ore-type condition

李路易

中国科学院数学与系统科学研究院

Let $n \in \mathbb{N}$ with $n \geq 3$, and let $\mathcal{G} = \{G_i : i \in [n]\}$ be a family of n -vertex graphs sharing the same vertex set V , where the graphs in the family need not be distinct. A graph H with vertex set V is *rainbow* in $\mathcal{G}$ if there exists an injection $\phi : E(H) \rightarrow [n]$ such that $e \in E(G_{\phi(e)})$ for every edge $e \in E(H)$, where $|E(G)| \leq n$. In 2020, Joos and Kim proved that $\mathcal{G}$ contains a rainbow Hamiltonian cycle under the Dirac-type condition. Recently, Liu, Chen, and Ma generalized this result by replacing the Dirac-type condition with a more general Ore-type condition involving degree sums of non-adjacent vertices: If $\sigma(\mathcal{G}) \geq n$, then $\mathcal{G}$ contains a rainbow Hamiltonian cycle, where the Ore-type condition $\sigma(\mathcal{G})$ is defined as follows:

$$\sigma(\mathcal{G}) = \min\{d_p(u) + d_q(v) \mid uv \notin E(G_i) \text{ for some } i \in [n] \text{ and for all } p, q \in [n]\}.$$

In this talk, under the Ore-type condition, we show that each vertex is contained in a rainbow cycle of length ℓ for each $\ell \in [4, n]$ in $\mathcal{G}$ or $G_1 = \cdots = G_n = K_{\frac{n}{2}, \frac{n}{2}}$. As a corollary, we deduce the rainbow pancyclicity of $\mathcal{G}$, which supports the famous meta-conjecture posed by Bondy. Furthermore, by a simple analysis, we obtain the rainbow vertex-pancyclicity of $\mathcal{G}$ under the Ore-type condition and give an extremal graph family to prove that this result is sharp.

个人简介：李路易，中国科学院数学与系统科学研究院博士后，合作导师为闫桂英教授，研究方向为结构图论。

Tight upper bound on the clique size in the square of 2-degenerate graphs

连晓盼

南开大学

The square of a graph G , denoted by G^2 , is obtained by joining every pair of vertices whose distance in G is at most two. The study of graph squares has attracted considerable attention in graph coloring.

For 2-degenerate graphs, it is known that the clique number of the square can be as large as $\frac{5}{2}\Delta(G)$, where $\Delta(G)$ denotes the maximum degree of G . This naturally raises the question of whether $\frac{5}{2}\Delta(G)$ is also an asymptotically sharp upper bound for the clique number of graph squares.

In this talk, we prove that there exists a constant D_0 such that $\omega(G^2) \leq \frac{5}{2}\Delta(G)$ whenever G is a 2-degenerate graph with $\Delta(G) \geq D_0$. Moreover, the bound is best possible.

个人简介：连晓盼，博士毕业于南开大学组合数学中心，导师为史永堂教授，现为
中心博士后，主要研究方向是图的染色问题。

A Sharp Product Bound for r -Cross-Intersecting Families

刘苗

山东大学-韩国基础科学研究院

We prove a sharp product theorem for r -cross-intersecting families in the p -biased measure. If $r \geq 2$, $0 \leq p \leq \frac{r-1}{r}$, and $\mathcal{F}_1, \dots, \mathcal{F}_r \subseteq 2^{[n]}$ are r -cross-intersecting, then

$$\prod_{i=1}^r \mu_p(\mathcal{F}_i) \leq p^r.$$

The bound is attained by a common 1-star, and the range of p is best possible. In particular, this proves the equal-bias case of a conjecture of Frankl and Tokushige and, for $r = 3$, confirms a conjecture of Tokushige.

For $r \geq 3$ we also prove a stability theorem: every near-extremal r -tuple is close, in p -biased measure, to a common 1-star. The extremal proof uses a coordinatewise coupling at the critical bias together with an isoperimetric inequality for increasing families. The stability proof uses dual families, a random ordered partition, and the biased Friedgut–Kalai–Naor theorem from analysis of Boolean functions. We also record a local extension for families satisfying the (k, q) -property.

个人简介：刘苗，山东大学-韩国基础科学研究院联合培养博士生，导师为上官冲教授（山东大学）和刘鸿教授（IBS），研究方向为极值组合与概率组合，编码理论。

A double-exponential lower bound for $r_4(5, n)$

刘睿龙

山东大学

The Ramsey number $r_k(s, n)$ is the smallest integer N such that every N -vertex k -graph contains either a copy of $K_s^{(k)}$ or an independent set of size n . We prove that $r_4(5, n) \geq 2^{2^{cn^{1/7}}}$, where $c > 0$ is an absolute constant. As a consequence, we determine the tower growth rate of $r_k(k+1, n)$, which completely solves the problem of establishing the tower growth rate for all classical off-diagonal hypergraph Ramsey numbers, first posed by Erdős and Hajnal in 1972.

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Fundamental cycles in grid graphs

舒希超

莱比锡大学

Given a connected graph G and a spanning tree T , every edge outside T gives rise to a unique cycle together with the path in T joining its endpoints; this cycle is called a fundamental cycle with respect to T . Fundamental cycles are classical objects in graph theory and naturally appear in the study of sparse representations of binary matroids. In particular, for graphic matroids, choosing a spanning tree as a basis yields an echelon-form representation in which the support sizes of the non-basis vectors correspond to the lengths of the associated fundamental cycles.

In this talk, I will discuss the average length of fundamental cycles in the $n \times n$ grid. We prove that, for every spanning tree of the grid, the average length of the fundamental cycles is at least $\Omega(\log n)$. This bound is asymptotically tight: we also construct a spanning tree for which the average fundamental cycle length is $O(\log n)$. Hence the optimal average length is $\Theta(\log n)$, answering a question of McCarty on sparse representations of binary matroids.

个人简介：舒希超，莱比锡大学博士生，导师 Dan Kral 教授（莱比锡大学及马克斯普朗克 MIS 研究所），研究方向为极值组合、结构图论以及离散几何。

Bound on shortest cycle covers

宋得萍

浙江师范大学

Assume G is a bridgeless graph. A cycle cover of G is a collection of cycles of G such that each edge of G is contained in at least one of the cycles. The length of a cycle cover of G is the sum of the lengths of the cycles in the cover. The minimum length of a cycle cover of G is denoted by $scc(G)$. The minimum length of a cycle cover of G consisting of k cycles is denoted by $scc_k(G)$. It was proved independently by Alon and Tarsi and by Bermond, Jackson, and Jaeger that $scc(G) \leq \frac{5}{3}m$ for every bridgeless graph G with m edges. This remained the best-known upper bound for $scc(G)$ for 40 years. In this paper, we prove that if G is a bridgeless graph with m edges and n_2 vertices of degree 2, then $scc_3(G) < \frac{29}{18}m + \frac{1}{18}n_2$. As a consequence, we show that $scc_3(G) \leq \frac{5}{3}m - \frac{1}{42}\log_2 m$. A key lemma used in the proof confirms Fan's conjecture that if C is a circuit of G and G/C admits a nowhere zero 4-flow, then G admits a 4-flow f such that $E(G) - E(C) \subseteq \text{supp}(f)$ and $|\text{supp}(f) \cap E(C)| > \frac{3}{4}|E(C)|$.

Theorem 1. *Let G be a bridgeless graph with m edges and n_2 vertices of degree 2, then $scc(G) < \frac{29}{18}m + \frac{1}{18}n_2$ and consequently $scc(G) \leq \frac{5}{3}m - \frac{1}{42}\log_2 m$.*

个人简介：宋得萍，浙江师范大学在读博士，导师为朱绪鼎教授。

Rational codegree Turán density of hypergraphs

孙淑敏

华威大学

Let H be a k -graph (i.e. a k -uniform hypergraph). Its *minimum codegree* $\delta_{k-1}(H)$ is the largest integer t such that every $(k-1)$ -subset of $V(H)$ is contained in at least t edges of H . The *codegree Turán density* $\gamma(\mathcal{F})$ of a family $\mathcal{F}$ of k -graphs is the infimum of $\gamma > 0$ such that every k -graph H on $n \rightarrow \infty$ vertices with $\delta_{k-1}(H) \geq (\gamma + o(1))n$ contains some member of $\mathcal{F}$ as a subgraph.

In this talk, we investigate the codegree Turán density of families of k -uniform hypergraphs. We present two main results. First, we prove a realizability result: for every integer k at least 3, every rational number in $[0, 1)$ is the codegree Turán density of some finite family of k -graphs. Second, we establish a strong version of non-principality. We construct two hypergraphs F_1 and F_2 such that the codegree Turán density of the pair $\{F_1, F_2\}$ is strictly smaller than the density of either F_1 or F_2 individually. This answers a question posed by Mubayi and Zhao (2007).

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On the structures of {diamond, bowtie}-free graphs that do not contain an induced subdivision of K_4

孙爽

上海交通大学

A graph is ISK_4 -free if it contains no induced subdivision of K_4 . In 2012, Lévêque, Maf-ray and Trotignon studied the structure of ISK_4 -free graphs and conjectured that every ISK_4 -free graph is 4-colorable. Chudnovsky, Liu, Schaudt, Spirk, Trotignon and Vušković proved that every $\{ISK_4, \text{triangle}\}$ -free graph is 3-colorable. Thus, a natural next step is to determine the optimal upper bound on the chromatic number of ISK_4 -free graphs that contain triangles.

Motivated by this, Chen, Chen, Cui, Feng and Liu considered the case where the interaction of triangles is restricted. In particular, they proved that every $\{ISK_4, \text{diamond}, \text{bowtie}\}$ -free graph is 4-colorable, where a diamond is K_4 minus one edge and a bowtie consists of two triangles sharing exactly one vertex. They further asked whether every graph in this class is in fact 3-colorable.

In this paper, we characterize the structure of $\{ISK_4, \text{diamond}, \text{bowtie}\}$ -free graphs and prove that every such graph is 3-colorable. This answers a question of Chen, Chen, Cui, Feng and Liu affirmatively and extends a result of Chudnovsky, Liu, Schaudt, Spirk, Trotignon and Vušković. Furthermore, our structural result is constructive. More precisely, for every $\{ISK_4, \text{diamond}, \text{bowtie}\}$ -free graph with n vertices and m edges, we give an algorithm that outputs such a decomposition in $O(n^2m)$ time. We also give an $O(n^2m)$ -time coloring algorithm that constructs a proper 3-coloring for every $\{ISK_4, \text{diamond}, \text{bowtie}\}$ -free graph.

个人简介：孙爽，上海交通大学在读博士，导师为汪彦副教授。

Multicolor Boolean Ramsey numbers

谭澳

南京师范大学

A Boolean lattice of dimension n , denoted Q_n , is the power set of an n -element ground set X equipped with inclusion relation. For posets $P_1, \dots, P_k$, the Boolean Ramsey number $R_k(P_1, \dots, P_k)$ is the smallest integer N such that for any k -coloring of the sets of Q_N , there exists a copy of P_i with color i . In this paper, we study the exact values or give upper and lower bounds of multicolor Boolean Ramsey numbers for some known posets and poset products.

个人简介：谭澳，南京师范大学在读博士，导师为张晓岩教授，研究方向为 Ramsey 理论。

A structure theorem for rooted connectivity in bidirected graphs

陶裘祯宇

华东理工大学

Recently, bidirected graphs have drawn increasing attention in graph theory, with fruitful progress on both structural and algorithmic aspects. They naturally generalize directed graphs and signed graphs: based on an undirected underlying graph, each endpoint of every edge is assigned a positive or negative sign. Compared with standard directed graphs, bidirected graphs possess much more intricate connectivity properties, which still remain insufficiently explored.

In this talk, I will show a structure theorem about rooted connectivity in bidirected graphs in terms of directed graphs. As applications, we prove Lovász's flame theorem, Pym's theorem and a strong variant of Menger's theorem for a class of bidirected graphs and provide counterexamples in the general case.

个人简介：陶裘祯宇，现任华东理工大学讲师。德国汉堡大学博士后，合作导师 Reinhard Diestel 与 Nathan Bowler 教授。博士毕业于南开大学组合数学中心，导师史永堂教授。研究方向有向图与无限图的结构问题。

Monochromatic Matchings and Tilings

王岚超

南京大学-韩国基础研究院

For a graph H and a host graph G , let $\Gamma_r(G)$ be the set of all r -edge-colourings of G . We define

$$Rt_r(H, G) := \min_{\chi \in \Gamma_r(G)} \nu_H(\chi),$$

where $\nu_H(\chi)$ is the maximum number of vertex-disjoint copies of H in a monochromatic H -tiling under the colouring χ . In the hypergraph matching setting, we write $\nu(\chi)$ for the maximum size of a monochromatic matching.

This talk presents a topology-free framework for monochromatic matchings and tilings, based on weak regularity and linear programming duality. It gives an elementary asymptotic approach to the Alon–Frankl–Lovász theorem and also leads to new results on graph Ramsey tilings.

Theorem 1. *Let $r, t \geq 2$, $\varepsilon > 0$ and $0 < d \leq 1 < D$. There exists $\eta > 0$ such that the following holds. Let $\mathcal{G}$ be a sufficiently large (η, p, d, D) -uniform t -graph on n vertices, equipped with an arbitrary r -edge-colouring χ . Then there exists an r -strip r -edge-colouring ψ of $K_n^{(t)}$ such that*

$$\nu(\chi) \geq \nu(\psi) - \varepsilon n.$$

Here an r -strip colouring means that the vertex set is partitioned into at most r parts, and the colour of an edge depends only on its intersection pattern with these parts.

Remark. This theorem reduces an arbitrary colouring of a pseudo-random t -uniform hypergraph to a highly structured strip colouring, while preserving the largest monochromatic matching up to $o(n)$. As a consequence, one obtains an elementary asymptotic proof of the Alon–Frankl–Lovász theorem,

$$\nu(\chi) \geq \frac{n}{r+t-1} - o(n),$$

as well as a self-contained proof of its random transference version.

Theorem 2. *Let $r \geq 2$ and let H be a connected graph with at least one edge. Then there exists a constant $\beta_{r,H}$, depending only on r and H , such that*

$$Rt_r(H, K_n) = (\beta_{r,H} + o(1))n.$$

Moreover, $\beta_{r,H}$ is effectively computable from a finite collection of linear programs whose sizes depend only on r and H .

Remark. This theorem asymptotically determines the multi-colour Ramsey tiling function $Rt_r(H, K_n)$ for every fixed graph H . It extends the classical two-colour tiling theorem of Burr, Erdős and Spencer to arbitrary numbers of colours. For $r \geq 3$, the result reveals a sharp structural dichotomy:

non-bipartite target graphs are governed by the trivial greedy bound, while bipartite target graphs require the LP-duality and strip-colouring framework.

个人简介：王岚超，南京大学-韩国基础研究院联合培养博士生，导师为陈耀俊教授和刘鸿教授，主要研究方向为 Ramsey 理论。

Color domination versus packing in graphs

郝常清

天津理工大学

The domination number and the packing number are two important parameters in graph theory. From a coloring perspective, this report introduces the concept of the k -color domination number and investigates its relationship with the packing number. The focus is on the 1-color domination number and the 2-color domination number, establishing upper bounds in terms of the packing number and the maximum degree of the graph. It is shown that, under given edge density conditions, the Cho-Kim Conjecture on the 1-color domination number being at most the maximum degree times the packing number holds for graphs with maximum degree at most 4, for 5-regular graphs, and for 6-regular graphs. Sharp bounds on the 2-color domination number in terms of the packing number are also established for graphs with maximum degree at most 4. Furthermore, a general proof framework and several structural lemmas applicable to the k -color domination number.

个人简介：郝常清，博士毕业于南开大学组合数学中心，导师为史永堂教授，现为天津理工大学数学科学学院讲师，研究方向为结构图论。

Triangle packings in randomly perturbed graphs

阎知非

韩国基础科学院

The longstanding Nash-Williams conjecture asserts that every K_3 -divisible graph G with $\delta(G) \geq 3n/4$ admits a triangle decomposition. In the random setting, Frankl and Rödl showed that, with high probability, $G(n, p)$ contains a triangle packing covering all but $o(n^2 p)$ edges whenever $p \geq n^{-1/2+\varepsilon}$.

In this talk, we consider near-perfect triangle packings in randomly perturbed graphs. We prove that for every $d > 0$ and every $p > 2d/(1 + 2d)$, if G_d is a dn -regular graph on n vertices, then with high probability the union $G_d \cup G(n, p)$ contains a triangle packing covering all but $o(n^2)$ edges. Moreover, this bound on p is best possible for $0 < d \leq 1/2$, thereby determining the threshold in this range. A key ingredient in the proof is a new triangle-weighting lemma for weighted complete graphs.

个人简介：阎知非，韩国基础科学院博后，博士毕业于 IMPA, 导师 Rob Morris。

Cayley Graph Chromatic Numbers and Applications

杨宁远

复旦大学

In this talk, we study the chromatic number of Cayley graphs over abelian groups, with a particular focus on the Hamming ball E_r of radius r around $\mathbf{1}$ in $\mathbb{F}_p^n$. We prove that for any prime p and integers n, r with $r \leq n/p$, the chromatic number of $\text{Cay}(\mathbb{F}_p^n, E_r)$ satisfies $\chi(\text{Cay}(\mathbb{F}_p^n, E_r)) \geq r/p^4$, establishing a quantitative lower bound for all primes p .

This result has two major applications bridging extremal combinatorics and dynamical systems. First, we use it to show that for every countably infinite abelian group Γ , there exists a set S that is topologically recurrent but not measurably recurrent, thereby separating classical recurrence properties in ergodic theory.

Second, we study the chromatic threshold $\delta_\chi(\mathcal{L})$ for homogeneous linear equations, generalizing Roth's theorem and recent Ramsey-Turán variants. We prove that $\delta_\chi(\mathcal{L}) = 0$ if and only if there exists a subset of coefficients of $\mathcal{L}$ of size at least three whose sum is zero.

To establish the chromatic lower bound, we introduce the ordered generalized Kneser graph and embed it into the Cayley graph structure. The bound then follows from an $\mathbb{F}_p$ -version of the Lyusternik-Shnirelman Theorem, utilizing topological tools akin to those in Lovász's and Greene's proofs of Kneser's conjecture.

个人简介：杨宁远，复旦大学博士生，导师为吴河辉教授。

Maximum size of digraphs with prescribed k -arc-strong connectivity

喻俊燃

南京师范大学

The k -arc-strong connectivity $\lambda'_k(D)$ of a strong digraph D , introduced by Day *et al.*, is the minimum number of arcs whose deletion produces at least k strong components. For integers n, k, t , let

$$F(n, k, t) = \max\{|A(D)| \mid |V(D)| = n, D \text{ is strong with } \lambda'_k(D) = t\}.$$

We determine this extremal function for all $3 \leq k < n$ and all possible values of t . More precisely, let $\lambda'_k(\overleftrightarrow{K}_n) = \varphi_k(n) = n(k-1) - \binom{k}{2}$, we prove the following theorem:

Theorem 1. *Let k, n, t be integers with $3 \leq k < n$ and $1 \leq t \leq n(k-1) - \binom{k}{2}$. Then*

$$F(n, k, t) = n(n-1) - n(k-1) + \binom{k}{2} + t.$$

Equivalently, $F(n, k, t) = n(n-1) - \varphi_k(n) + t$.

The proof consists of a general upper bound, and shows that the bound is sharp by giving a extremal construction. This completely determines the maximum size of digraphs for k -arc-strong connectivity.

个人简介：喻俊燃，南京师范大学在读博士，导师为张晓岩教授，研究方向为图论与组合优化。

Degree stability of graphs forbidding a family of odd cycles

袁小利

湖南大学

In this talk, we will share some recent progress on degree stability of graphs forbidding a family of odd cycles. This is joint work with Yuejian Peng and Zilong Yan.

个人简介：袁小利，湖南大学在读博士生，导师为彭岳建教授，研究方向为极值图论。

Minimum spectral radii of saturated graphs for linear forests

张俊雪

北京理工大学

A graph G is called H -saturated if G does not contain H as a subgraph, but the addition of any missing edge creates a copy of H . We write $\text{sat}_\rho(n, H)$ for the minimum spectral radius among all n -vertex H -saturated graphs.

This talk concerns spectral saturation problems for the linear forests tP_3 and tP_4 . We determine the exact values of $\text{sat}_\rho(n, tP_3)$ and $\text{sat}_\rho(n, tP_4)$, and characterize all extremal graphs attaining these values. We also compare these spectral extremal results with the classical saturation problem, showing that, for tP_4 with $t \geq 3$ and $n \geq 6t + 4$, the graphs minimizing the spectral radius are different from those minimizing the number of edges. The proofs are based on structural analysis of saturated graphs and spectral comparison arguments.

个人简介：张俊雪，北京理工大学博士后，合作导师：韩杰教授，研究方向为结构图论和极值图论。

An Ore-type condition for H -tilings in graphs

张林朋

山东大学

A graph G admits an H -tiling if it contains a collection of vertex-disjoint copies of H . In this paper, we confirm a conjecture proposed by Kühn, Osthus, and Treglown by showing that for any given graph H , there exists a constant $C(H)$ such that the following holds. If G is a sufficiently large n -vertex graph satisfying $d(x) + d(y) \geq 2(1 - 1/\chi_{\text{cr}}(H))n$ for all nonadjacent vertices $x, y \in V(G)$, then G contains an H -tiling covering all but at most $C(H)$ vertices. Here $\chi_{\text{cr}}(H)$ denotes the critical chromatic number of H .

个人简介：张林朋，山东大学数学学院博士后，合作导师：王光辉教授，博士毕业于西北工业大学和荷兰特文特大学，主要研究方向为图和超图中子图存在性问题。

Hamilton cycles in oriented graphs with mild connectivity constraints

张灵浩

山东大学

We investigate Hamilton cycles in regular oriented graphs under the mild connectivity assumption of strong well-connectedness and strongly 3 and 4-connectivity. Our work is motivated by directed analogues of the Bollobás–Häggkvist Conjecture and by the questions of Lo, Patel and Yıldız, who asked whether every strongly well-connected d -regular oriented graph on n vertices with $d \geq n/6$ must be Hamiltonian and whether every strongly 3-connected d -regular oriented graph on n vertices with $d \geq n/8$ must be Hamiltonian. We answer those questions in the negative. We prove sharp Hamiltonicity thresholds for regular oriented graphs under those mild connectivity assumption, and we construct matching extremal examples showing that these degree conditions are best possible.

Definition 1. A digraph G on at least four vertices is strongly well-connected if for every partition (A, B) of $V(G)$ with $|A|, |B| \geq 2$, there exist two vertex-disjoint edges ab and cd such that $a, d \in A, b, c \in B$.

Theorem 2. Let G be a d -regular oriented graph on n vertices with n is sufficiently large. Suppose that one of the following holds:

- G is strongly well-connected and $d \geq (3n - 4)/14$;
- G is strongly 3-connected and $d \geq (3n - 5)/14$;
- G is strongly 4-connected and $d \geq (3n - 6)/14$.

Then G has a Hamilton cycle.

个人简介：张灵浩，山东大学数学学院在读硕士，导师为王光辉教授，研究方向为极值图论。

Perfect Matchings in Random Sparsifications of Dense Hypergraphs

赵婧雯

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Given $1 \leq \ell < k$ and $\delta \geq 0$, let $\mathbf{PM}(k, \ell, \delta)$ be the decision problem for the existence of perfect matchings in n -vertex k -uniform hypergraphs with minimum ℓ -degree at least $\delta \binom{n-\ell}{k-\ell}$. For $k \geq 3$, $\mathbf{PM}(k, \ell, 0)$ was one of the first NP-complete problems identified by Karp. Keevash, Knox and Mycroft conjectured that $\mathbf{PM}(k, \ell, \delta)$ is in P for every $\delta > 1 - (1 - 1/k)^{k-\ell}$ and this was recently verified for $\ell \geq 0.4k$ by Gan and Han.

In this paper we study the existence of perfect matchings in the random p -sparsification of such k -uniform hypergraphs, that is, for $p = p(n) \in [0, 1]$, every edge is kept with probability p independent of others. Building on the structural theory of Gan and Han, we show that the corresponding dense perfect matching results are robust under random sparsification. As a consequence, we obtain polynomial-time algorithms that asymptotically almost surely solve the associated decision problems, as well as lower bounds on the number of perfect matchings in such hypergraphs. We also establish analogous results for the F -factor problem in graphs.

Our proofs combine the lattice-based absorption method with the random redistribution method of Kelly, Mueyesser and Pokrovskiy based on spread distributions.

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Sufficient conditions for fractional k -factor-critical graphs with minimum degree to be k -factor-critical

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A graph G is called k -factor-critical if after deleting any k vertices the remaining subgraph still has a perfect matching. A graph G is fractional k -factor-critical if after deleting any k vertices the remaining subgraph still has a fractional perfect matching. Clearly, the fractional k -factor-criticality of a graph is a necessary property for a graph to be k -factor-critical. We first establish a lower bound on the size to ensure that a $(k+1)$ -connected graph with fractional k -factor-criticality is k -factor-critical, where k is a positive integer with $k \geq 1$. Moreover, we provide a sufficient condition in terms of the spectral radius for a $(k+1)$ -connected graph with fractional k -factor-criticality to be k -factor-critical.

Theorem 1. *Let G be a $(k+1)$ -connected graph with fractional k -factor-criticality of order $n \geq \max\{6\delta - 5k + 8, \frac{\delta^2}{6} - (\frac{k}{3} - \frac{3}{2})\delta + \frac{k^2}{6} - \frac{k}{2} + 4\}$, where δ is the minimum degree of G , $k \geq 1$ is an integer and $n \equiv k \pmod{2}$. If*

$$|E(G)| \geq |E(K_\delta \vee (K_{n-2\delta+k-3} \cup K_3 \cup (\delta-k)K_1))|,$$

then G is k -factor-critical unless $G \cong K_\delta \vee (K_{n-2\delta+k-3} \cup K_3 \cup (\delta-k)K_1)$.

Theorem 2. *Let G be a $(k+1)$ -connected graph with fractional k -factor-criticality of order $n \geq \max\{8\delta - 5k + 20, \delta(\delta - k - 1)^2 - 2\delta - 1\}$, where δ is the minimum degree of G , $k \geq 1$ is an integer and $n \equiv k \pmod{2}$. If*

$$\rho(G) \geq \rho(K_\delta \vee (K_{n-2\delta+k-3} \cup K_3 \cup (\delta-k)K_1)),$$

then G is k -factor-critical unless $G \cong K_\delta \vee (K_{n-2\delta+k-3} \cup K_3 \cup (\delta-k)K_1)$.

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Fractional balanced chromatic number and arboricity of planar graphs

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A *signed graph* (G, σ) is a graph $G = (V, E)$ together with an assignment $\sigma : E(G) \rightarrow \{+, -\}$. Sign of a substructure in (G, σ) is the product of the signs of edge in the structure. A set of vertices is said to be *balanced* if it induces no negative cycle.

A *balanced (p, q) -coloring* of (G, σ) is an assignment of q colors to each vertex of G from a platter of p colors, such that each colors class induces a balanced set. The *fractional balanced chromatic number* of a signed graph (G, σ) is defined as

$$\chi_{fb}(G, \sigma) = \min\left\{\frac{p}{q} : (G, \sigma) \text{ admits a } (p, q)\text{-coloring}\right\}.$$

Fractional balanced chromatic number of a signed graph (G, σ) is upper-bounded by the fractional arboricity of G . *Fractional arboricity* of G , denoted $a_f(G)$, is the minimum of $\frac{p}{q}$ such that one can assign (at least) q colors to each vertex using only a total of (at most) p colors such that each color class is acyclic.

Albertson and Berman conjectured that every planar graph G has an acyclic set of order at least $\frac{|V(G)|}{2}$. Bonamy, Kardoš, Kelly, and Postle proposed a strengthening of the Albertson–Berman conjecture: every planar graph admits a set of induced forests such that each vertex appears in at least half of them. That is equivalent to:

Conjecture 1. *Every planar graph G satisfies $a_f(G) \leq 2$.*

Using the signed version of Wenger graph as a key gadget, we build a planar signed (simple) graph for which $\chi_{fb}(G, \sigma) > 2$. The underlying graph of this construction then is a counterexample to Conjecture 1.

Theorem 1. *The supremum of the fractional balanced chromatic number of planar signed simple graphs is at least $2 + \frac{1}{41}$.*

Theorem 2. *There exists a planar graph G with $a_f(G) = 2 + \frac{2}{25}$.*

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Ore-type degree condition for Hamiltonicity and pancyclicity of hypergraphs and bipartite graphs

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Let $\mathcal{H}$ be a 2-connected hypergraph on $n \geq 3$ vertices. A Hamilton Berge-cycle in $\mathcal{H}$ consists of a cyclic ordering $v_1, \dots, v_n$ of all vertices in $V(\mathcal{H})$ together with n distinct edges $e_1, \dots, e_n \in E(\mathcal{H})$ such that $\{v_i, v_{i+1}\} \subseteq e_i$ for all i , with indices taken modulo n (i.e., $v_{n+1} = v_1$). The hypergraph $\mathcal{H}$ is super-pancyclic if for every subset $A \subseteq V(\mathcal{H})$ with $|A| \geq 3$, there is a Berge-cycle whose set of base vertices is exactly A . Recently, Kostochka, Luo, and Zirlin [J. Comb. Theory, Ser. B 145 (2020) 450-465] gave a tight Dirac-type degree condition for a 2-connected hypergraph $\mathcal{H}$ to be Hamiltonian and super-pancyclic. In this paper, we extend this result to a tight Ore-type degree condition.

Observe that a sequence $C = v_1 e_1 v_2 e_2 \dots v_n e_n v_1$ is a Hamilton Berge-cycle of $\mathcal{H}$ if and only if it corresponds to a cycle in the incidence graph $I(\mathcal{H})$ that covers $V(\mathcal{H})$. Based on this correspondence, our proofs are carried out mostly in the language of bipartite graphs.

Theorem 1.5. Every 2-connected bipartite graph $G = (X, Y; E) \in \mathcal{G}(n, m, \sigma_2)$ with $\sigma_2 \geq \max\{2n, \frac{2m+9}{3}\}$ contains an X -covering cycle.

Theorem 1.7. Let $G = (X, Y; E) \in \mathcal{G}(n, m, \sigma_2)$ be a graph with $\sigma_2 \geq \max\{2n, \frac{2m+9}{3}\}$. If G satisfies the condition (NC), then G is X -super-cyclic.

Remark. The bound $\frac{2m+9}{3}$ in Theorem 1.5 is sharp. Moreover, these theorems strictly generalize the previous Dirac-type results of Kostochka, Luo and Zirlin, as they apply to graphs that do not meet the minimum degree condition.

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